AutonomouStuff touts its "completely customisable R&D vehicle platforms used for ADAS, advanced algorithm development and automated driving initiatives." Conventional ADAS technology can detect some objects, do basic classification, alert the driver of hazardous road conditions and, in some cases, slow or stop the vehicle. This level of ADAS is great for applications like blind spot monitoring, lane-change assistance and forward collision warnings.

NVIDIA DRIVE PX 2 AI car computers take driver assistance to the next level. These take advantage of deep learning and include a software development kit (SDK) for autonomous driving called Drive Works. This SDK gives developers a powerful foundation for building applications that leverage computation-intensive algorithms for object detection, map localisation and path planning. With NVIDIA self-driving car solutions, a vehicle's ADAS can discern a police car from a taxi, an ambulance from a delivery truck, or a parked car from one that is about to pull out into traffic. It can even extend this capability to identify everything from cyclists on the sidewalk to absent-minded pedestrians.

OpenPilot. OpenPilot project consists of two component parts: on-board firmware and ground control station (GCS). The firmware is written in 'C,' whilst ground control station is written in 'C++' utilising Qt. This platform is meant for development only.

OpenPilot is basically a behaviour model based on Comma.ai's trained network. Comma.ai, a startup, used quick hacking of the car's network bus to simplify having the computer control the car. They did it almost entirely with CNNs. The car feeds images from a camera into the network, and out from the network come commands to adjust the steering and speed to keep a car in its lane. As such, there is very little traditional code in the system, just the neural network and a bit of control logic.

The network is built by training it. As a car is driven around, it learns from the human driving what to do when it sees things in the field of view. Light Detection and Ranging (LIDAR) gives the car an accurate 3D scan of the environment to more absolutely detect the presence of cars and other users of the road. By getting to know during training that there is really something there at these coordinates, the network can learn how to tell the same thing from just the camera images.

When it is time to drive, the network does not get the LIDAR data, however it does produce outputs of where it thinks the other cars are, allowing developers to test how well it is seeing things. This allows development of a credible autopilot, but, at the same time, developers have minimal information about how it works, and can never truly understand why it is making the decisions it does. If it makes an error, they will generally not know why it made the error, though they can give it more training data until it no longer makes the error.

Developing an autonomous car requires a lot of training data. To drive using vision, developers need to segment 2D images from cameras into independent objects to create a 3D map of the scene. Other than using parallax, relative motion and two cameras to do that, autonomous cars also need to see patterns in the shapes to classify them when they see them from every distance and angle, and in every lighting condition. That's been one of the big challenges for vision—you must understand things in sunlight, in diffuse light, in night-time (when illuminated by headlights, for example) when the sun is pointed directly into the camera, when complex objects like trees are casting moving shadows on the desired objects and so on. You must be able to identify pavement marking, lane marking, shoulders, debris, other road users, etc in all these situations and you must do it so well that you make a mistake perhaps only once every million miles. A million miles is around 2.7 billion frames of video.

Autonomous driving startups

The goal is to build a hardware and software kit powered by artificial intelligence for carmakers. Drive.ai, an AI software startup with expertise in robotics and deep learning, applies machine learning to both driving and human interaction.

Startup FiveAI develops a host of technologies including sensor fusion, computer vision combined with a deep neural network, policy-based behavioural modelling and motion planning. The goal is to design solutions for a Level 5 autonomous vehicle that is safe in complex urban environments.

Nauto offers an AI-powered dual-camera unit that learns from drivers and the road. Using real-time sensors and visual data, it focuses on insight, guidance and real-time feedback for its clients, helping fleet managers detect and understand causes of accidents and reduce false liability claims. The system also helps cities with better traffic control and street design to eliminate fatal accidents.

Oxbotica specialises in mobile autonomy, navigation, perception and research into autonomous robotics. It has been engaged in the development of self-driving features for some time, and purports to train their algorithms using AI for complex urban environments by directly mapping sensor data against actual driving behav-